

EC3452 EMF Lesson Plan – Schedule

Here's a complete Lesson Plan for EC3452 – Electromagnetic Fields (Anna University Regulation 2021, IV Semester ECE):

This plan is structured unit-wise and week-by-week, aligned with the official syllabus [A Plus Topper](https://www.auniversity.ac.in/eeeonline.org) [eeeonline.org](https://www.auniversity.ac.in/eeeonline.org) [Inspire n Ignite \(InI\)](https://www.auniversity.ac.in/eeeonline.org).

Course Overview

- **Course Code:** EC3452
- **Course Title:** Electromagnetic Fields
- **Regulation:** Anna University R2021
- **Semester:** IV (ECE)
- **Credits:** 3 (Lecture-based)
- **Course Objectives:**
 - Understand static electric and magnetic fields.
 - Learn vector calculus and coordinate systems.
 - Study Maxwell's equations and time-varying fields.
 - Analyze EM wave propagation in different media.
 - Apply concepts to transmission lines and practical systems.

Lesson Plan (15 Weeks)

Week	Topics	Teaching Methods	Learning Outcomes
1	Introduction – Vector algebra, coordinate systems, gradient, divergence, curl	Lecture + Tutorial	Apply vector calculus to EM problems
2	Vector Theorems – Divergence theorem, Stokes theorem, Helmholtz theorem	Problem-solving	Verify vector identities in EM fields
3	Electrostatics I – Coulomb's law, electric field intensity, field due to point/line/plane charges	Lecture + Examples	Calculate E-field for charge distributions

4	Electrostatics II – Electric flux density, Gauss’s law, applications	Tutorial + Assignments	Apply Gauss’s law to symmetric charge systems
5	Electrostatics III – Electric potential, potential gradient, boundary conditions	Lecture + Problem-solving	Relate potential and field intensity
6	Conductors & Dielectrics – Capacitance, Poisson’s & Laplace’s equations	Simulation + Tutorial	Solve boundary-value problems
7	Magnetostatics I – Biot-Savart law, Ampere’s law, magnetic field intensity	Lecture + Worked examples	Compute H-field for current distributions
8	Magnetostatics II – Magnetic flux density, Gauss’s law for magnetism	Tutorial	Apply laws to magnetic circuits
9	Magnetostatics III – Vector potential, inductance, boundary conditions	Lecture + Assignments	Derive inductance for simple geometries
10	Time-Varying Fields I – Faraday’s law, displacement current	Lecture + Simulation	Understand coupling of E and H fields
11	Time-Varying Fields II – Maxwell’s equations (differential & integral forms)	Problem-solving	Apply Maxwell’s equations to EM systems
12	EM Waves I – Wave equation, plane waves in free space	Lecture + Tutorial	Derive wave propagation equations
13	EM Waves II – Waves in dielectrics, conductors, skin depth	Simulation + Lab demo	Analyze wave behavior in different media
14	Transmission Lines – Parameters, wave propagation, reflection, VSWR	Lecture + Problem-solving	Apply EM concepts to transmission lines
15	Revision & Assessment – Mixed problems, model exam	Quiz + Group discussion	Integrate concepts across all units

Teaching Strategies

- **Lectures:** Core theory with derivations
- **Tutorials:** Step-by-step problem solving
- **Simulations:** MATLAB/COMSOL for EM wave visualization
- **Assignments:** Weekly problem sets
- **Quizzes:** Short tests for continuous assessment

Evaluation Pattern

- **Internal Assessment:** 40 marks (Assignments, quizzes, tutorials)
- **End Semester Exam:** 60 marks (3-hour written exam)

Suggested Textbooks

- Matthew N.O. Sadiku, *Elements of Electromagnetics*
- William H. Hayt & John A. Buck, *Engineering Electromagnetics*
- David K. Cheng, *Field and Wave Electromagnetics*

✅ This lesson plan ensures **systematic coverage of all five units** in EC3452, balancing theory, problem-solving, and practical exposure.

Would you like me to also prepare a **weekly assignment schedule (problem sets + simulation tasks)** starting from a specific date, similar to what we did for Circuit & Network Analysis?

Sources: [A Plus Topper](#) [eeeonline.org](#) [Inspire n Ignite \(InI\)](#)

Great! Let's design a **Weekly Assignment Schedule** for **EC3452 – Electromagnetic Fields (Anna University R2021, IV Semester ECE)** starting from **12 January 2026 (Monday)**.

This schedule spans **15 weeks** and integrates both **problem sets** and **simulation tasks** to reinforce theory with practice.

Weekly Assignment Schedule (12.01.2026 – 26.04.2026)

Week	Date Range	Assignment Focus	Problem Set	Simulation Task
1	Jan 12 – Jan 18	Vector Calculus Basics	Solve 10 problems on gradient, divergence, curl	MATLAB: Plot vector fields in Cartesian coordinates

2	Jan 19 – Jan 25	Vector Theorems	Apply Divergence & Stokes theorem to 5 problems	MATLAB: Verify divergence theorem numerically
3	Jan 26 – Feb 1	Electrostatics I	Coulomb's law problems for point/line charges	COMSOL: Electric field distribution of point charge
4	Feb 2 – Feb 8	Electrostatics II	Gauss's law problems for sphere, cylinder	MATLAB: Flux calculation using surface integrals
5	Feb 9 – Feb 15	Electrostatics III	Potential & boundary condition problems	COMSOL: Potential distribution between parallel plates
6	Feb 16 – Feb 22	Conductors & Dielectrics	Capacitance problems + Laplace equation	MATLAB: Capacitance calculation for coaxial cable
7	Feb 23 – Mar 1	Magnetostatics I	Biot-Savart law problems for line/current loop	MATLAB: Plot H-field around a current-carrying wire
8	Mar 2 – Mar 8	Magnetostatics II	Ampere's law problems for solenoid, toroid	COMSOL: Magnetic field inside a solenoid
9	Mar 9 – Mar 15	Magnetostatics III	Inductance calculation problems	MATLAB: Inductance of coaxial cable
10	Mar 16 – Mar 22	Time-Varying Fields I	Faraday's law & displacement current problems	MATLAB: Induced emf in a rotating loop
11	Mar 23 – Mar 29	Maxwell's Equations	Derive & apply Maxwell's equations to 3 cases	COMSOL: Simulate E & H field coupling
12	Mar 30 – Apr 5	EM Waves I	Wave equation problems in free space	MATLAB: Plane wave propagation visualization
13	Apr 6 – Apr 12	EM Waves II	Skin depth & wave in conductors problems	COMSOL: Wave attenuation in lossy medium
14	Apr 13 – Apr 19	Transmission Lines	Reflection coefficient, VSWR problems	MATLAB: Transmission line simulation with mismatched load
15	Apr 20 – Apr 26	Revision & Assessment	Mixed problems from all units	MATLAB/COMSOL: Comprehensive EM

				wave + transmission line demo
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■ Unit I – Vector Calculus & Electrostatics

Q: State Coulomb's law.

A: Force between two point charges (q_1, q_2) separated by distance (r): $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$

Q: Define divergence.

A: Divergence of vector field ($\vec{A}$) is ($\nabla \cdot \vec{A}$), representing net flux per unit volume.

Q: Find electric field at 1 m from a charge of ($2 \mu\text{C}$).

A:

$$(E = \frac{q}{4\pi\epsilon_0 r^2} = \frac{2 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 1^2} \approx 1.8 \times 10^5, \text{ V/m}).$$

Q: State Gauss's law.

A: Total electric flux through closed surface equals charge enclosed divided by permittivity:

$$\left(\oint \vec{D} \cdot d\vec{S} = Q_{\text{enclosed}} \right).$$

Q: Define electric potential.

A: Work done per unit charge in moving a test charge from infinity to a point in the field.

■ Unit II – Conductors, Dielectrics & Capacitance

Q: Define capacitance.

A: Ratio of charge stored to potential difference: ($C = Q/V$).

Q: Find capacitance of parallel plate capacitor with area (1 m^2), separation (0.01 m), air dielectric.

$$\text{A: } (C = \frac{\epsilon_0 A}{d} = 8.85 \times 10^{-12} \times 1/0.01 = 8.85 \times 10^{-10} \text{ F}).$$

Q: State boundary condition for electric field at conductor surface.

A: Tangential component of ($\vec{E}$) is zero; normal component equals surface charge density divided by (ϵ).

Q: Define dielectric strength.

A: Maximum electric field a dielectric can withstand without breakdown.

Q: Write Poisson's equation.

A: $\left(\nabla^2 V = -\frac{\rho}{\epsilon}\right).$

■ Unit III – Magnetostatics

Q: State Biot-Savart law.

A: Magnetic field due to current element:

$$(d\vec{H} = \frac{I d\vec{l} \times \hat{r}}{4\pi r^2})$$

Q: Find magnetic field at 0.1 m from a long wire carrying 10 A.

A: $\left(H = \frac{I}{2\pi r} = \frac{10}{2\pi \times 0.1} \approx 15.9, \frac{A}{m}\right).$

Q: State Ampere's law.

A: Line integral of $(\vec{H})$ around closed path equals current enclosed:

$$\left(\oint \vec{H} \cdot d\vec{l} = I\right).$$

Q: Define magnetic flux density.

A: $(\vec{B} = \mu\vec{H})$, measured in Tesla.

Q: Define inductance.

A: Ratio of flux linkage to current: $(L = N\Phi/I)$.

■ Unit IV – Time-Varying Fields & Maxwell's Equations

Q: State Faraday's law.

A: Induced emf equals negative rate of change of magnetic flux: $(e = -\frac{d\Phi}{dt})$.

Q: Define displacement current.

A: Current equivalent due to time-varying electric field: $(I_d = \epsilon \frac{d\Phi_E}{dt})$.

Q: Write Maxwell's curl equation for $(\vec{E})$. **A:** $(\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t})$.

Q: Find induced emf when flux changes from 0.02 Wb to 0.01 Wb in 0.1 s.

A: $\left(e = -\frac{\Delta\Phi}{\Delta t} = -\frac{0.01 - 0.02}{0.1} = 0.1 V\right).$

Q: Define conduction current density.

A: ($J = \sigma E$), where (σ) is conductivity.

■ Unit V – Electromagnetic Waves & Transmission Lines

Q: Write wave equation in free space.

A: $\left(\vec{\nabla}^2\{E\} = \mu_0\epsilon_0 \frac{\partial^2\{E\}}{\partial t^2}\right).$

Q: Define skin depth.

A: Depth at which wave amplitude reduces to $\left(\frac{1}{e}\right)$ of surface value: $\left(\delta = \sqrt{\frac{2}{\omega\mu\sigma}}\right).$

Q: Find wavelength in free space at 1 GHz.

A: $\left(\lambda = \frac{c}{f} = 3 \times \frac{10^8}{10^9} = 0.3 \text{ m}\right).$

Q: Define VSWR.

A: Ratio of maximum to minimum voltage on transmission line: $VSWR = \left(\frac{1+|\Gamma|}{1-|\Gamma|}\right).$

Q: State condition for lossless transmission lines.

A: Resistance ($R = 0$) and conductance ($G = 0$).